



Original article

Cooperative guidance under multiple constraints without active speed control

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ABSTRACT

To address the challenge of coordinated strikes by multiple missiles without active speed control, a three-dimensional cooperative guidance law is proposed, based on a second-order system consensus protocol that incorporates multiple constraints. The law accounts for field-of-view limitations, input saturation, and no-fly zones. To eliminate errors caused by estimating time-to-go, an auxiliary system is introduced. A fuzzy rule-based phased cooperative guidance law is then proposed to balance strike accuracy and coordination consistency. Additionally, a behavior-based obstacle avoidance method, based on the interfered fluid dynamical system algorithm, is designed to enhance the aircraft's survivability. The convergence properties of the proposed algorithm are rigorously proven using Lyapunov theory. Numerical and Monte Carlo simulations demonstrate its effectiveness and superiority across various scenarios.

1. Introduction

Advancements in air defense technologies have substantially diminished the combat effectiveness of individual aircraft. However, cooperative operations involving multiple aircraft can effectively mitigate the limitations of one-on-one engagements, enhancing both survivability and overall combat capability. The simultaneous launch of multiple missiles from different locations to strike a target has garnered significant attention due to its potential to enhance combat effectiveness [1].

Jeon et al. [2] first introduced the concept of attack-time control guidance, enabling salvo attacks by multiple missiles. A common method is to adjust the arrival times of the aircraft individually. A phased cooperative guidance law is proposed for hypersonic gliding vehicles in [3]. In the first stage, the attack time is coordinated by adjusting the bank angle and angle of attack, while a cooperative terminal guidance law is developed for the driving-down phase, considering the terminal attack time and angle. This method achieves time coordination among multiple aircraft by independently adjusting the time for each aircraft to reach the target, effectively transforming the multi-to-one time-coordinated strike problem into a time adjustment issue for a single aircraft [4,5].

Another method involves the implementation of multi-agent consensus protocols, which realizes true time-coordinated guidance through

information exchange and decision making [6]. For the problem of coordinated strike by multiple missiles against stationary targets, based on a multi-agent consensus protocol, a distributed cooperative guidance law is designed and further extended to scenarios with communication failures in [7]. For missiles with constant speed, Kong et al. [8] designed a fixed-time convergent cooperative guidance law, considering the constraint on the attack angle. A cooperative guidance law with a biased proportional navigation form is proposed to achieve synchronized strike timing, where strike time is determined by the maximum consensus algorithm without prior specification. An equivalent verification platform is built to further validate the effectiveness of the algorithm in [9].

However, the aforementioned articles require the estimation of time-to-go and rely on the consensus protocol to achieve the synchronization of the remaining time. The estimation of time-to-go is based on the small-angle assumption. When there is a large leading angle between the aircraft and the target, the small-angle assumption no longer holds, resulting in significant estimation errors, which in turn affect the coordination performance [10–12]. In recent years, cooperative guidance without using the time-to-go has become a research hotspot. Chunyan et al. [13] designed a phased cooperative guidance law, where the first stage ensures synchronized strike timing without using the time-to-go information, while the second stage ensures the minimization of miss distance. Based on a second-order system consensus protocol, Zhao

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et al. [14] designed a fixed-time convergent sliding mode guidance law and achieved effective avoidance of multiple no-fly zones.

Additionally, to facilitate the practical application of the cooperative guidance law to aircraft, it is essential to consider various constraints, such as field-of-view (FOV) limitations, input saturation, and no-fly zones.

The onboard computer needs to calculate guidance commands based on the missile information measured by the seeker. Due to the missile's large maneuvering, the limited FOV of the seeker may cause it to lose track of the target, leading to mission failure. Sang and Tahk [15] first considered the FOV constraint of the aircraft in the cooperative guidance law and proposed a switching logic to maintain the FOV angle. Yang et al. [16] proposed a two-stage cooperative guidance law for stationary targets with impact angle and FOV constraints. The FOV constraints are ensured by a multi-agent consensus algorithm with constraints and a leading angle tracking theory.

However, the aforementioned methods did not consider actuator saturation constraints in their design, which could result in system instability. In practical systems, it is necessary to take into account the saturation constraints of the actuator to ensure that the guidance commands can be executed effectively. Yang et al. [17] proposed a cooperative guidance law under asymmetric input constraints, where a Tanh function was designed to ensure that the aircraft satisfies the asymmetric input constraints. For the problem of multiple missiles cooperatively attacking a maneuvering target, Wang et al. [18] proposed a novel predetermined-time cooperative guidance law and introduced a first-order auxiliary system to address the issue of input saturation.

In addition to the constraints imposed by the aircraft itself, the aircraft must also adhere to environmental constraints. To ensure its safety, it is necessary for the aircraft to avoid certain no-fly zones during the strike process. Zhao et al. [14] proposed a spatiotemporal cooperative guidance law based on a second-order consensus system and developed a no-fly zones avoidance algorithm using the interfered fluid dynamical system (IFDS) method and the singularity avoidance problem is further considered.

However, the aforementioned studies are based on the assumption that the aircraft's speed is either constant or continuously adjustable. It is worth noting that most aircraft are propelled by rocket engines with predetermined thrust, indicating that active speed control is largely infeasible in practice. Furthermore, the velocity is influenced by a combination of engine thrust, gravity, and air resistance, which conflicts with the assumption of constant speed. Therefore, developing cooperative guidance laws for missiles without active speed control under multiple constraints holds significant practical relevance.

For the challenge of a single unpowered gliding vehicle guidance under impact velocity and impact angle constraints, Kong et al. [8] proposed a multi-constraint guidance law based on fixed-time convergent error dynamics. Dong et al. [19] adopted a progressive design strategy to achieve three-dimensional (3D) spatiotemporal cooperative guidance for multiple missiles with time-varying speeds. They also considered the seeker's potential field constraints and designed a bias term to ensure that the leading angle does not exceed its upper limit. However, these references do not consider no-fly zones constraints and input saturation constraints.

Therefore, from the above discussion it is evident that time-coordinated guidance for aircraft without active speed control under input saturation constraints, FOV constraints, and no-fly zone constraints remains an open and challenging problem. To overcome these challenges, this paper proposes a cooperative guidance law for aircraft without active speed control, designed to operate under complex constraints. The main contributions of this work are as follows:

1. Compared with Zhao et al. [14], the constraint that the aircraft's speed cannot be actively controlled is considered. And the time-coordinated guidance law design does not rely on the estimation of time-to-go. Instead, the leading angle, relative distance, and closing velocity are used as coordination variables.

2. Compared with Dong et al. [19], for the cooperative guidance of vehicles without active speed control, this approach considers not only the FOV constraint but also the input constraint and no-fly zones constraint.
3. Taking into account both coordination accuracy and strike effect, this paper proposes a fuzzy rule based cooperative guidance law, which alleviate the impact of instruction shock caused by phase switching in [13]. Additionally, the problem of no-fly zones avoidance is further considered, and a behavior-based no-fly zones avoidance method is proposed.

The remainder of this paper is structured as follows. Section 2 introduces the necessary preliminaries and relevant kinematic equations. Section 3 presents the cooperative guidance law and obstacle avoidance method, along with a theoretical analysis of its convergence characteristics. In Section 4, simulations and comparative analysis are conducted to demonstrate the effectiveness and improvements of the proposed algorithm. Finally, Section 5 presents the conclusions.

2. Problem statement and preliminaries

2.1. Problem statement

This section presents the kinematic model for cooperative strike by multiple aircraft without active speed control. The time-coordinated guidance law proposed in this paper is designed to enable coordinated strikes, taking into account constraints such as FOV limitations, input saturation, and no-fly zones.

Consider the cooperative strike in 3D space against a stationary or slow-moving target. Fig. 1 illustrates the cooperative attack scenario considering no-fly zones, where $O_L x_L y_L z_L$ represents the inertial coordinate system, θ_L, φ_L denotes the elevation and azimuth angle of the LOS, respectively. θ_m, φ_m denotes the elevation and azimuth angle of the leading angle, respectively. δ means the total leading angle. According to the geometric relationship, $\delta = \arccos(\cos \theta_m \cos \varphi_m)$.

The 3D relative kinematics and dynamics model can be expressed as [20,21]:

$$\begin{cases} \dot{r}_i = -V_{m,i} \cos \theta_{m,i} \cos \varphi_{m,i} \\ r_i \dot{\theta}_{L,i} = -V_{m,i} \sin \theta_{m,i} \\ r_i \dot{\varphi}_{L,i} \cos \theta_{L,i} = -V_{m,i} \cos \theta_{m,i} \sin \varphi_{m,i} \\ \dot{\theta}_{m,i} = \frac{a_{y,i}}{V_{m,i}} - \dot{\varphi}_{L,i} \sin \theta_{L,i} \sin \varphi_{m,i} - \dot{\theta}_{L,i} \cos \varphi_{m,i} \\ \dot{\varphi}_{m,i} = \frac{a_{z,i}}{V_{m,i} \cos \theta_{m,i}} + \dot{\varphi}_{L,i} \tan \theta_{m,i} \cos \varphi_{m,i} \sin \theta_{L,i} \\ \quad - \dot{\theta}_{L,i} \tan \theta_{m,i} \sin \varphi_{m,i} - \dot{\varphi}_{L,i} \cos \theta_{L,i} \\ \dot{V}_{m,i} = \frac{T_i - D_i}{m_i} - g_i \sin \theta_i \end{cases} \quad (1)$$

where r_i is the distance between the target and the aircraft, $V_{m,i}$ is the aircraft's velocity, $a_{y,i}$ and $a_{z,i}$ are the accelerations of the aircraft along the Y-axis and Z-axis in the velocity coordinate system. T_i is the engine thrust, ignoring the angle of attack. D_i is the aerodynamic drag. g_i represents the gravitational acceleration. m_i represents the mass of the aircraft. θ_i is the aircraft's flight path angle.

Remark 1. The drag D_i is determined by the angle of attack, Mach, and altitude. As the angle of attack increases, the corresponding drag also increases. Therefore, as in reference [20,21], it can be approximated that the rate of change in the aircraft's speed due to the aerodynamic drag D_i is positively correlated with the required acceleration of the aircraft:

$$\dot{V}_{m,i} = T_i/m_i - g_i \sin \theta_i - k_{v,i} |a_i| + \Delta V_{m,i}$$

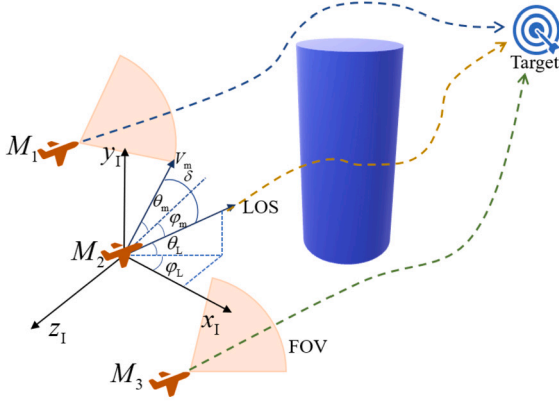


Fig. 1. Cooperative guidance for aircraft without active speed control.

$$= T_i/m_i - g_i \sin \theta_i - k_{v,i} \sqrt{a_{y,i}^2 + a_{z,i}^2} + \Delta V_{m,i} \quad (2)$$

where $k_{v,i}$ represents the rate coefficient, which is related to the aircraft's aerodynamic coefficients and mass, and $\Delta V_{m,i}$ represents the unmodeled errors and other disturbances.

2.2. Preliminaries

The no-fly zone can be constructed as:

$$\Gamma(\mathbf{X}) = \left(\frac{x - x_0}{a} \right)^{2p} + \left(\frac{y - y_0}{b} \right)^{2q} + \left(\frac{z - z_0}{c} \right)^{2r} \quad (3)$$

where $\mathbf{X} = [x, y, z]^T$ represents the position of the aircraft, $[x_0, y_0, z_0]^T$ is the center of the no-fly zone. Parameters $a, b, c > 0$ determine the size of the no-fly zone and $p, q, r > 0$ determine the shape [22].

Lemma 1 ([23]). Consider the velocity of the flow field in a 3D scenario as $\mathbf{u}(x, y, z)$. When there is an obstacle as shown in Eq. (3), the flow field becomes $\tilde{\mathbf{u}}(x, y, z)$. Let

$$\zeta = [x - x_0, y - y_0, z - z_0]^T \quad (4)$$

For a point outside the obstacle, the modulation matrix can be defined as:

$$\mathbf{M} = \mathbf{I} - \frac{\mathbf{n}\mathbf{n}^T}{|\zeta|^\rho \mathbf{n}^T \mathbf{n}} \quad (5)$$

where $\mathbf{I}$ is the unit matrix, ρ is a non-negative hyperparameter, $\mathbf{n} = \left[\frac{\partial \Gamma}{\partial x}, \frac{\partial \Gamma}{\partial y}, \frac{\partial \Gamma}{\partial z} \right]^T$.

Then the velocity of the flow field affected by the obstacle can be expressed as:

$$\tilde{\mathbf{u}}(x, y, z) = \mathbf{M}(x, y, z)\mathbf{u}(x, y, z) \quad (6)$$

The flow field has the following properties:

1. The obstacle is impenetrable, that is, when the point is on the surface of the obstacle, the normal velocity is 0.
2. The disturbance caused by the obstacle tends to zero as the distance approaches infinity.

Assuming a sink flow is located at the origin, with the origin positioned outside the obstacle, the flow field $\tilde{\mathbf{u}}$ exhibits the following properties:

1. Outside the obstacle, there is always a non-zero component pointing to the origin, that is, the total velocity is always toward the target point.
2. The origin is the only singular point.

The flow field distribution within the domain is calculated by hydrodynamics, and the streamlines in the flow field are utilized as the flight trajectories for the aircraft. The target point is treated as the origin of the sink flow. The above characteristics of the disturbed flow field ensure the feasibility of the obstacle avoidance algorithm.

Consider the aircraft group with n aircraft, the communication topology is modeled by the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$, where $\mathcal{V} = \{v_1, v_2, v_3, \dots\}$ denotes the set of all nodes, $\mathcal{E} = \{(i, j) \subseteq \mathcal{V} \times \mathcal{V}\}$ is the edges, and $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{n \times n}$ is the adjacency matrix. If $(j, i) \in \mathcal{E}$, then $a_{ij} = 1$; otherwise, $a_{ij} = 0$, with the specific condition that $a_{ii} = 0$. The Laplacian matrix $\mathcal{L}$ is defined as $\mathcal{L}_{ii} = \sum_{j \neq i} a_{ij}$ and $\mathcal{L}_{ij} = -a_{ij}, i \neq j$.

Lemma 2 ([24]). Consider a strongly connected second-order multi-agent system with n individuals,

$$\begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = u_i \end{cases} \quad (7)$$

A distributed consensus controller with both velocity and input constraints can be designed as:

$$\begin{aligned} u_i &= -K(v_i - v_i^r) \\ v_i^r &= v_a - B \tanh \left[K \sum_{j=1}^n a_{ij}(x_i - x_j) + \sum_{j=1}^n a_{ij}(v_i - v_j) \right] \end{aligned} \quad (8)$$

where K is the hyperparameter of the controller, $K > 0$. $v_a = \frac{1}{2}(v_{\min} + v_{\max})$, $B = \frac{1}{2}(v_{\max} - v_{\min})$, where $v_{\max}$ and $v_{\min}$ are the maximum and minimum values of state v_i respectively.

3. Fuzzy rule based cooperative guidance law

In this section, for missiles without active speed control, a 3D cooperative guidance law is developed to achieve temporal attack under FOV constraint, input saturation, incorporating no-fly zones avoidance. By ensuring the convergence of the auxiliary state variables, the attack time can be guaranteed to be consensus from an analytical perspective without using the estimate of time-to-go.

3.1. Design of cooperative guidance law with FOV and input constraints

The nonlinear nature of the system described in Eq. (1) presents significant challenges in guidance law design. To address this complexity, an auxiliary system is introduced to facilitate the transformation of the nonlinear system into a linear formulation.

3.1.1. Construction of auxiliary systems

The auxiliary variables are designed as:

$$\begin{cases} \eta_i = \frac{r_i}{V_{m,i}} \\ \xi_i = -\cos \delta_i \end{cases} \quad (9)$$

Remark 2. Considering the limited FOV angle of the missile:

$$\delta_{\min} \leq \delta_i \leq \delta_{\max} \quad (10)$$

where $\delta_{\min}$ and $\delta_{\max}$ are the lower and upper limits of the FOV angle.

According to Eq. (9), the FOV constraint can be transformed into the auxiliary state constraint, that is, $\xi_{\min} \leq \xi_i \leq \xi_{\max}$, where $\xi_{\min}$ and $\xi_{\max}$ are the boundaries of the auxiliary state ξ_i . Therefore, the auxiliary system Eq. (9) is a second-order system with velocity limitation.

By taking the derivative of Eq. (9) and coupling it with Eq. (1), it can be obtained that:

$$\begin{cases} \dot{\eta}_i = \frac{\dot{r}_i}{V_{m,i}} - \frac{r_i \dot{V}_{m,i}}{V_{m,i}^2} \\ \dot{\xi}_i = \hat{u}_i \end{cases} \quad (11)$$

where

$$\begin{aligned} \hat{u}_i = & \frac{a_{z,i}}{V_{m,i}} \sin \varphi_{m,i} + \frac{a_{y,i}}{V_{m,i}} \sin \theta_{m,i} \cos \varphi_{m,i} + \frac{V_{m,i}}{r_i} \sin^2 \theta_{m,i} \\ & + \frac{V_{m,i}}{r_i} \cos^2 \theta_{m,i} \sin^2 \varphi_{m,i} \end{aligned} \quad (12)$$

In the design of the consensus protocol, the velocity $V_{m,i}$ is considered constant, which means that the velocity variation $\dot{V}_{m,i}$ is treated as a small quantity and can be neglected. Let $\tilde{u}_i = a_{y,i} \sin \theta_{m,i} \cos \varphi_{m,i} + a_{z,i} \sin \varphi_{m,i}$. Considering $\dot{r}_i = -V_{m,i} \cos \delta_i$, thus Eq. (11) can be further simplified to:

$$\begin{cases} \dot{\eta}_i = \xi_i \\ \dot{\xi}_i = \frac{\tilde{u}_i}{V_{m,i}} + \frac{V_{m,i}}{r_i} \sin^2 \theta_{m,i} + \frac{V_{m,i}}{r_i} \cos^2 \theta_{m,i} \sin^2 \varphi_{m,i} + d_i \end{cases} \quad (13)$$

In Eq. (13), since the effect of $\frac{r_i \dot{V}_{m,i}}{V_{m,i}^2}$ is ignored, the term d_i represents the disturbance caused by the velocity variations to the system, and this term is bounded. The influence of this disturbance is ignored in the guidance law design.

Therefore, the original nonlinear control problem is converted into a problem of using the new control $\hat{u}_i$ to control a second-order linear system with disturbances. At the same time, considering that the leading angle of the aircraft δ_i and the control input $\hat{u}_i$ are bounded, so Eq. (13) is a consistency problem with bounded velocity and input.

3.1.2. Design of cooperative guidance law and command solution

According to the Lemma 2, the controller is designed as:

$$\begin{cases} \hat{u}_i = -k_1(\xi_i - \xi_i^r) \\ \xi_i^r = \xi_a - b \tanh \left[k_1 \sum_{j=1}^n a_{ij}(\eta_i - \eta_j) + \sum_{j=1}^n a_{ij}(\xi_i - \xi_j) \right] \end{cases} \quad (14)$$

where $\xi_a = \frac{1}{2}(\xi_{\min} + \xi_{\max})$, $b = \frac{1}{2}(\xi_{\max} - \xi_{\min})$, and $k_1 > 0$.

Theorem 1. *With respect to the n strongly connected second-order multi-agent systems as shown in Eq. (13), the controller provided in Eq. (14) guarantees bounded inputs and velocities, ensuring consensus convergence of the multi-agent system.*

The following proof is divided into two parts. First, it is proved that when the initial conditions meet the constraints, the auxiliary system state ξ_i and the control instructions $\hat{u}$ will not exceed the boundary. Next, it is proved that the controller is effective and can ensure the consistent convergence of the second-order system.

Proof. Given that the $\tanh()$ function is confined to the range $[-1, 1]$, it follows directly from Eq. (14) that $\xi_{\min} \leq \xi_i^r \leq \xi_{\max}$.

If the initial conditions meet the constraints, that is, $\xi_{\min} \leq \xi_i(0) \leq \xi_{\max}$. The closed set $D := \{\xi_i \in \mathbb{R} \mid \xi_{\min} \leq \xi_i \leq \xi_{\max}\}$ is considered. To establish the positive invariance of the set D , the behavior of the system on the boundaries of D is analyzed:

1) When $\xi_i = \xi_{\max}$, since $\xi_i^r \leq \xi_{\max}$, it follows that $\xi_i - \xi_i^r \geq 0$, and thus $\hat{u}_i = -k_1(\xi_i - \xi_i^r) \leq 0$. Therefore, the state cannot increase beyond $\xi_{\max}$ at the upper boundary.

2) When $\xi_i = \xi_{\min}$, since $\xi_i^r \geq \xi_{\min}$, it follows that $\xi_i - \xi_i^r \leq 0$, and thus $\hat{u}_i = -k_1(\xi_i - \xi_i^r) \geq 0$. Hence, the state cannot decrease below $\xi_{\min}$ at the lower boundary.

Since the vector field always points inward or is tangential on the boundary of D , and assuming the initial condition satisfies the con-

straints, $\xi(0) \in D$, it follows from the Nagumo's theorem that $\xi_i(t) \in D$ for all $t \geq 0$.

Since both ξ_i and ξ_i^r are bounded, that is, $\xi_{\min} \leq \xi_i, \xi_i^r \leq \xi_{\max}$. Therefore, for the controller designed in Eq. (14), it is obvious that,

$$|\hat{u}_i| \leq k_1 |\xi_{\max} - \xi_{\min}| \quad (15)$$

Therefore, when $k_1 \leq \frac{u_{\max}}{\xi_{\max} - \xi_{\min}}$, it can be ensured that $|\hat{u}_i| \leq u_{\max}$, where $u_{\max}$ denotes the upper limit of the acceleration.

Next, it is proved that with the distributed controller Eq. (14), consensus can be achieved.

Let

$$\begin{cases} e_i^\eta = \sum_{j=1}^n a_{ij}(\eta_i - \eta_j) \\ e_i^\xi = \sum_{j=1}^n a_{ij}(\xi_i - \xi_j) \end{cases} \quad (16)$$

Then the distributed controller can be rewritten as:

$$\hat{u}_i = -k_1 [\xi_i - \xi_a + b \tanh(k_1 e_i^\eta + e_i^\xi)] \quad (17)$$

By taking the derivative of Eq. (16), it can be obtained that,

$$\begin{aligned} \dot{e}_i^\xi &= \sum_{j=1}^n a_{ij} (\hat{u}_i - \hat{u}_j) \\ &= -k_1 b \sum_{i=1}^n a_{ij} \left[\tanh(k_1 e_i^\eta + e_i^\xi) - \tanh(k_1 e_j^\eta + e_j^\xi) \right] - k_1 e_i^\xi \end{aligned} \quad (18)$$

Denote $\mathbf{e}^\eta = [e_1^\eta, e_2^\eta, \dots, e_n^\eta]^T$, $\mathbf{e}^\xi = [e_1^\xi, e_2^\xi, \dots, e_n^\xi]^T$, then

$$\begin{cases} \dot{\mathbf{e}}^\eta = \mathbf{e}^\xi \\ \dot{\mathbf{e}}^\xi = -(\mathbf{I}_N \otimes k_1) \mathbf{e}^\xi - (\mathcal{L} \otimes k_1 b) \tanh[(\mathbf{I}_N \otimes k_1) \mathbf{e}^\eta + \mathbf{e}^\xi] \end{cases} \quad (19)$$

where $\mathcal{L}$ is the Laplacian matrix of graph $\mathcal{G}$, $\mathbf{I}_N$ is the n -dimensional identity matrix.

Let $\zeta = \mathbf{e}^\xi + (\mathbf{I}_N \otimes k_1) \mathbf{e}^\eta$, then

$$\dot{\zeta} = -(\mathcal{L} \otimes k_1 b) \tanh(\zeta) \quad (20)$$

If the graph $\mathcal{G}$ is strongly connected, there exists a positive-definite matrix $\mathbf{P} = \text{diag}\{p_1, p_2, \dots, p_n\}$ such that $\mathbf{Q} = \mathbf{P}\mathcal{L} + \mathcal{L}^T\mathbf{P}$ is positive semidefinite [25]. Choose the following Lyapunov function,

$$V(\zeta) = \sum_{i=1}^n p_i \int_0^{\zeta_i} \tanh(s) ds \quad (21)$$

Since $\tanh()$ is an odd function, V is positive definite. And the derivative of V is

$$\begin{aligned} \dot{V}(\zeta) &= -\tanh^T(\zeta)(\mathbf{P}\mathcal{L} \otimes k_1 b) \tanh(\zeta) \\ &= -\frac{1}{2} \tanh^T(\zeta)(\mathbf{Q} \otimes k_1 b) \tanh(\zeta) \\ &\leq 0 \end{aligned} \quad (22)$$

According to the properties of the $\tanh()$ function, the equality $\dot{V}(\zeta) = 0$ holds if and only if $\zeta = 0$. Therefore, the system is stable, which means $\zeta = \mathbf{e}^\xi + (\mathbf{I}_N \otimes k_1) \mathbf{e}^\eta = 0$, where $\dot{\mathbf{e}}^\eta = \mathbf{e}^\xi$ and $k_1 > 0$. Thus $\mathbf{e}^\eta \rightarrow 0$, $\mathbf{e}^\xi \rightarrow 0$, which means when $t \rightarrow \infty$, $\eta_i \rightarrow \eta_j$, $\xi_i \rightarrow \xi_j$.

So far, it has been proved that the controller Eq. (14) can ensure that the FOV of the aircraft does not exceed the boundary, the input of the controller has an upper bound, and it can ensure that multiple aircraft are coordinated under strong connectivity.

After obtaining the expression for $\tilde{u}_i$ from the controller, it is necessary to further derive the expressions for $a_{y,i}$ and $a_{z,i}$. In this paper, the expressions for $a_{y,i}$ and $a_{z,i}$ are obtained based on the principle of minimizing the total acceleration [26]. That is

$$\min a_i = \sqrt{a_{y,i}^2 + a_{z,i}^2} \quad (23)$$

$$s.t. \quad \tilde{u}_i = a_{y,i} \sin \theta_{m,i} \cos \varphi_{m,i} + a_{z,i}$$

Therefore, it can be accurately solved that,

$$\begin{cases} a_{y,i} = \frac{\sin \theta_{m,i} \cos \varphi_{m,i} (\hat{u}_i - \frac{V_{m,i}^2}{r_i} \sin^2 \theta_{m,i} - \frac{V_{m,i}^2}{r_i} \cos^2 \theta_{m,i} \sin^2 \varphi_{m,i})}{\sin^2 \theta_{m,i} \cos^2 \varphi_{m,i} + \sin^2 \varphi_{m,i}} \\ a_{z,i} = \frac{\sin \varphi_{m,i} (\hat{u}_i - \frac{V_{m,i}^2}{r_i} \sin^2 \theta_{m,i} - \frac{V_{m,i}^2}{r_i} \cos^2 \theta_{m,i} \sin^2 \varphi_{m,i})}{\sin^2 \theta_{m,i} \cos^2 \varphi_{m,i} + \sin^2 \varphi_{m,i}} \end{cases} \quad (24)$$

Considering the saturation of the actuator, the acceleration command is:

$$\begin{cases} \bar{a}_{y,i} = \text{Sat}(a_{y,i}) \\ \bar{a}_{z,i} = \text{Sat}(a_{z,i}) \end{cases} \quad (25)$$

where $\text{Sat}()$ is the saturation function, specifically,

$$\text{Sat}(x) = \begin{cases} x_{\max} & \text{if } x > x_{\max} \\ x & \text{if } x_{\min} \leq x \leq x_{\max} \\ x_{\min} & \text{if } x < x_{\min} \end{cases} \quad (26)$$

Remark 3. According Eq. (22), the hyperparameter k_1 determines the controller convergence speed, and also determines the upper limit of control. Although the upper limit of control has no direct correspondence with the actual aircraft control input $a_{y,i}$ and $a_{z,i}$, it is still necessary to design k_1 according to the aircraft's capabilities to avoid the aircraft being in input saturation for a long time.

3.2. Proportional navigation guidance (PNG) law

In the design of the cooperative guidance law, the auxiliary state ξ_i is designed to converge to a fixed reference value, meaning that the terminal leading angle is not zero, which may lead to significant terminal errors. To ensure accurate target interception during the terminal phase, it is necessary to reduce the total leading angle to zero. Therefore, in the terminal phase of guidance, the guidance law switches to PNG to ensure guidance accuracy.

Theorem 2 ([13]). When applying the PNG law, the time required for the aircraft to reach the target depends solely on the terminal values of the auxiliary system η_i and ξ_i .

Proof. When considering the plane of the aircraft's trajectory, the 3D relative kinematic trajectory can be simplified to planar motion within the trajectory profile.

$$\begin{cases} \dot{r}_i = -V_{m,i} \cos \delta_i \\ \dot{\lambda}_{L,i} = -\frac{V_{m,i} \sin \delta_i}{r_i} \\ \dot{\gamma}_i = \frac{a_i}{V_{m,i}} \end{cases} \quad (27)$$

where $\lambda_{L,i}$ is the LOS elevation angle, γ_i is the flight path angle. a_i is the acceleration command, when using the PNG guidance law, $a_i = NV_{m,i} \dot{\lambda}_{L,i}$, where N is the PNG coefficient.

Then it can be derived that,

$$\begin{aligned} \dot{\delta}_i &= \dot{\gamma}_i - \dot{\lambda}_{L,i} \\ &= -\frac{(N-1)V_{m,i} \sin \delta_i}{r_i} \end{aligned} \quad (28)$$

The time for the aircraft to reach the target can be expressed as:

$$\begin{aligned} t_{f,i} - t_{0,i} &= \int_{r_{0,i}}^0 \frac{dt}{dr} dr \\ &= \frac{1}{V_{m,i}} \int_0^{r_{0,i}} \sec \delta_i dr \\ &= \frac{r_{0,i}}{V_{m,i}} \int_0^1 \sec \delta_i d\tau \end{aligned} \quad (29)$$

where $\tau = \frac{r_i}{r_{0,i}}$. Thus

$$\dot{\tau} = -\frac{V_{m,i} \cos \delta_i}{r_{0,i}} \quad (30)$$

Combine Eq. (30) with Eq. (28), it can be drawn that,

$$\frac{d\delta_i}{d\tau} = \frac{(N-1) \tan \delta_i}{\tau} \quad (31)$$

which means $\int_0^1 \sec \delta_i d\tau$ depends solely on the initial state of the leading angle. Therefore, when using a guidance law with the same proportional coefficient, the final time for the aircraft to reach the target depends only on the terminal values of the auxiliary system $r_{0,i}/V_{m,i}$ and δ_i .

Remark 4. When the auxiliary states η_i and ξ_i converge to consensus, the time for the aircraft to reach the target region after switching to PNG will also remain consistent, that is, $t_{f,i} = t_{f,j}$, $\forall i, j$. This demonstrates that PNG exhibits a bounded error propagation property, inheriting only the initial error $e(0)$ at the switching instant, and satisfying $\|e(t_f)\| \leq \|e(0)\|$, thereby avoiding the introduction of additional errors.

Consequently, a fuzzy switching system is designed to adaptively adjust the weight coefficients of the cooperative guidance law and PNG based on the missiles' cooperative state and the distance between the aircraft and the target, as shown in Fig. 2. When the coordination error is significant, the cooperative guidance law takes precedence, achieving rapid synchronization of cooperative states via the distributed consensus protocol. As the coordination error diminishes, the weight of PNG progressively increases to optimize strike accuracy. This approach not only mitigates the transient disturbances associated with guidance law switching but also ensures the cooperative term persistently attenuates residual errors, thereby averting the adverse effects of prematurely terminating the cooperative guidance law.

Furthermore, when the distance between the aircraft and the target is less than R_a , the guidance law will fully switch to the PNG guidance law to ensure the final guidance accuracy.

Definition 1. Define the cooperative state as C_{oo} to measure the consistency between aircraft,

$$C_{oo} = \beta_1 \sum_{i=1}^n (\eta_i - \bar{\eta}) + \beta_2 \sum_{i=1}^n (\xi_i - \bar{\xi}) \quad (32)$$

where $\bar{\eta} = \frac{1}{n} \sum_{i=1}^n \eta_i$, $\bar{\xi} = \frac{1}{n} \sum_{i=1}^n \xi_i$. β_1, β_2 represent the weights assigned to the auxiliary states, respectively.

And the average distance $\bar{D}$ between the aircraft group and the target is used as the variable to measure the distance between the aircraft and the target, where

$$\bar{D} = \sum_{i=1}^n r_i / n \quad (33)$$

The fuzzy switching logic takes the cooperative state C_{oo} and the distance $\bar{D}$ as inputs, and the weight coefficients of the cooperative guidance law and PNG guidance law as outputs. The input domain is

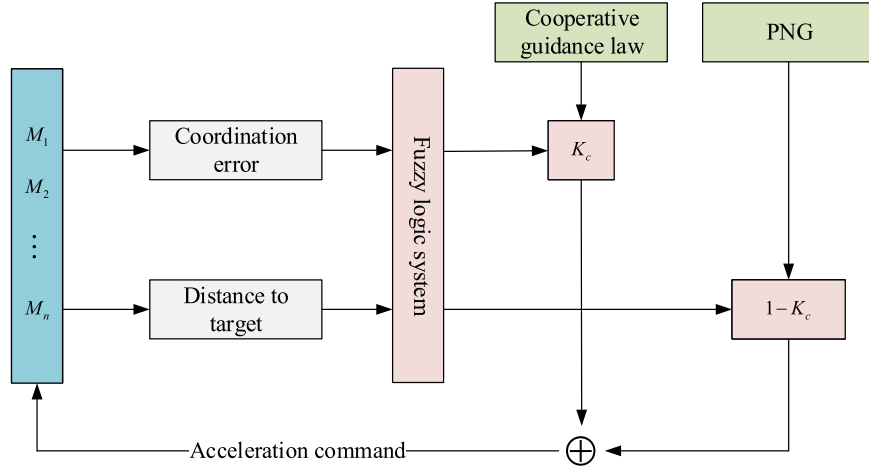


Fig. 2. Fuzzy logic based compound guidance.

Table 1
Fuzzy rules base.

	$\bar{D}$		
C_{oo}	N	ZO	P
N	N	N	N
ZO	N	ZO	P
P	N	ZO	P

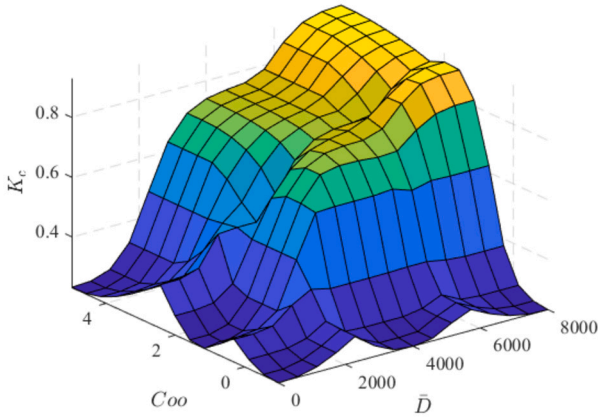


Fig. 3. Fuzzy rule surface.

divided into three levels: N (negative), ZO (zero), and P (positive), and the membership functions of the fuzzy subsets on the domain are Gaussian membership functions. The output domain is also divided into three levels: N (negative), ZO (zero), and P (positive). The inference algorithm uses the Mamdani inference mechanism, and the defuzzification method employs the centroid method. The fuzzy rule base is shown in Table 1, and the fuzzy rule surface is shown in Fig. 3.

3.3. Behavior based no-fly zones avoidance

The initial velocity flow field of the aircraft is defined as:

$$\mathbf{u}_i(\mathbf{X}_i) = - \left(\frac{V_{m,i} \Delta x_i}{d(\mathbf{X}_i, \mathbf{P}_d)}, \frac{V_{m,i} \Delta y_i}{d(\mathbf{X}_i, \mathbf{P}_d)}, \frac{V_{m,i} \Delta z_i}{d(\mathbf{X}_i, \mathbf{P}_d)} \right)^T \quad (34)$$

where $\mathbf{X}_i = [x_{m,i}, y_{m,i}, z_{m,i}]^T$ represents the position of the aircraft, $\mathbf{P}_d = [x_d, y_d, z_d]^T$ denotes the position of the target. $\Delta x_i = x_{m,i} - x_d$, $\Delta y_i = y_{m,i} - y_d$, $\Delta z_i = z_{m,i} - z_d$. And $d()$ denotes the distance between the two points.

According the Lemma 1, the interfered fluid velocity can be expressed as

$$\bar{\mathbf{u}}_i(\mathbf{X}_i) = \mathbf{M}_i(\mathbf{X}_i) \mathbf{u}_i(\mathbf{X}_i) \quad (35)$$

Thus, the expected position at the next time step can be expressed as:

$$\mathbf{P}_{t+1,i} = \mathbf{P}_{t,i} + \bar{\mathbf{u}}_i(\mathbf{X}_i) \Delta T \quad (36)$$

where ΔT is the prediction step size. Properly increasing the prediction step size ΔT can smooth the flight trajectory and command acceleration.

After obtaining the desired position, PNG is used to follow the target:

$$\begin{cases} a_{y,i} = N V_{m,i} \dot{\gamma}_{L,i} \\ a_{z,i} = N V_{m,i} \dot{\psi}_{L,i} \end{cases} \quad (37)$$

where $\gamma_{L,i}$, $\psi_{L,i}$ denote the LOS elevation angle and the LOS azimuth angle between the aircraft and the target $\mathbf{P}_{t+1,i}$, respectively.

When the aircraft enters a certain range of the no-fly zone, which is described as the decision area, the guidance law switches to the obstacle avoidance algorithm, and when the aircraft leaves the range, it switches back to the above cooperative guidance law. Furthermore, it is worth noting that, since the FOV limitation is not considered in the obstacle avoidance process, when the aircraft leaves the decision area, initial alignment is required to ensure that the initial state of the collaborative guidance meets the constraints, as shown in Fig. 4. During the initial alignment phase, the leading angle between the target and the aircraft is first calculated. If the leading angle does not satisfy the predefined constraints, a reverse control force is actively applied to adjust the attitude of the vehicle, ensuring that the leading angle falls within the permissible range, thereby completing the initial alignment.

In summary, the complete algorithm flowchart for the cooperative guidance law of aircraft without active speed control, considering FOV constraints, input constraints, and no-fly zone restrictions, is shown in Fig. 5.

4. Simulations

Different from the assumption in theoretical analysis that the velocity variation can be ignored, in the simulation environment, the velocity of the missile is affected by gravity, thrust and air drag. To further verify the effectiveness and robustness of the proposed method, numerical simulation analysis is carried out under different scenarios. The initial conditions and hyper-parameters are provided in Table 2. Meanwhile, the missiles initially point at the target, with leading angles of zero, meeting the constraints.

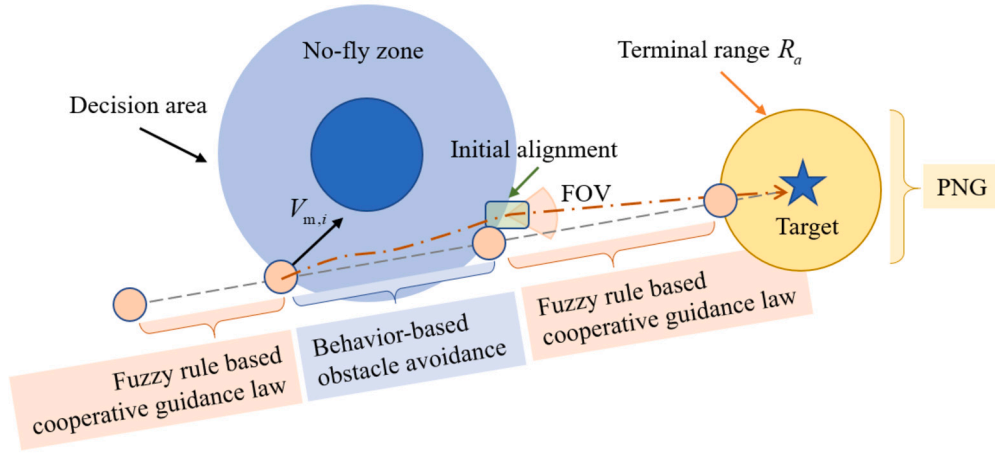


Fig. 4. Phased cooperative guidance law.

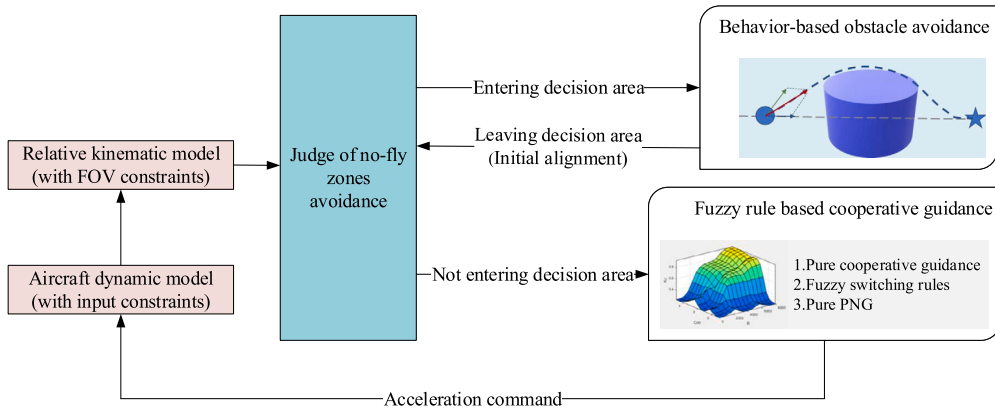


Fig. 5. Flowchart of the proposed cooperative guidance law.

Subsequently, the results are compared and analyzed, followed by a Monte Carlo experiment to assess the stability of the algorithm.

4.1. Numerical simulation

Multiple sets of comparative simulations are conducted to validate the advancements of the proposed algorithm.

Case 1. Cooperation without no-fly zones avoidance.

The no-fly zone avoidance is not considered in this scenario, focusing on validating the effectiveness of the cooperative guidance law for multi-aircraft coordination without active speed control, while addressing FOV constraints and input saturation.

Fig. 6(a) illustrates the 3D trajectories of four aircraft during cooperative guidance. Fig. 6(b) shows the distances between each aircraft and the target, with the maximum miss distance reaching 5.54 m at the end of the simulation. Meanwhile, the cooperative system is affected by unmodeled disturbances, which are mainly caused by velocity variations, resulting in incomplete convergence of the missiles' arrival times, with terminal time errors within 0.5 s. Without considering no-fly zones, the aircraft penetrate the restricted area, and the distance between the aircraft and the center of the no-fly zone is presented in Fig. 6(c).

During the cooperative phase, shown as Fig. 6(d), the auxiliary variables η_i and ξ_i converge consistently. The introduction of the PNG law drives the total leading angle $\delta_i \rightarrow 0$, which means $\xi_i \rightarrow -1$. The fuzzy rule design enables a linear combination of the cooperative guidance law and the PNG law, resulting in the auxiliary variable ξ_i exhibiting both convergence and a tendency toward -1 .

Definition 2. The coordination error e_{coo} is defined as the standard deviation of the arrival time of each aircraft when the simulation stops:

$$e_{\text{coo}} = \sqrt{\frac{\sum_{i=1}^n (T_i - \bar{T})^2}{n}} \quad (38)$$

where T_i is the arrival time of i -th missile, and $\bar{T} = \frac{1}{n} \sum_{i=1}^n T_i$. Taking into account the missiles' velocity and simulation step size, the simulation stops when $r_i \leq 5$ m.

Definition 3. To facilitate comparative analysis of convergence rates, the auxiliary variable η is selected for detailed examination. Inspired by the statistical concept of the coefficient of variation, which quantifies data dispersion, the convergence time T_{con} is defined as the time it takes for the standard deviation of the auxiliary variable η to converge to within 3% of its mean [27].

$$T_{\text{con}} : \sqrt{\frac{\sum_{i=1}^n (\eta_i - \bar{\eta})^2}{n}} \leq 3\% \bar{\eta} \quad (39)$$

where $\bar{\eta} = \frac{1}{n} \sum_{i=1}^n \eta_i$.

When the no-fly zone is not considered, the convergence time of the proposed algorithm is 15.30 s, with a coordination error of 0.43 s.

Fig. 6(e) illustrates the acceleration profiles along the y and z -axes. In the initial stage of simulation, the instructions reach saturation and then gradually decrease. When the average distance of the aircraft group from the target exceeds 3 Km, a linear weighting of the cooperative guidance law and PNG is applied, where the weighting coefficient of

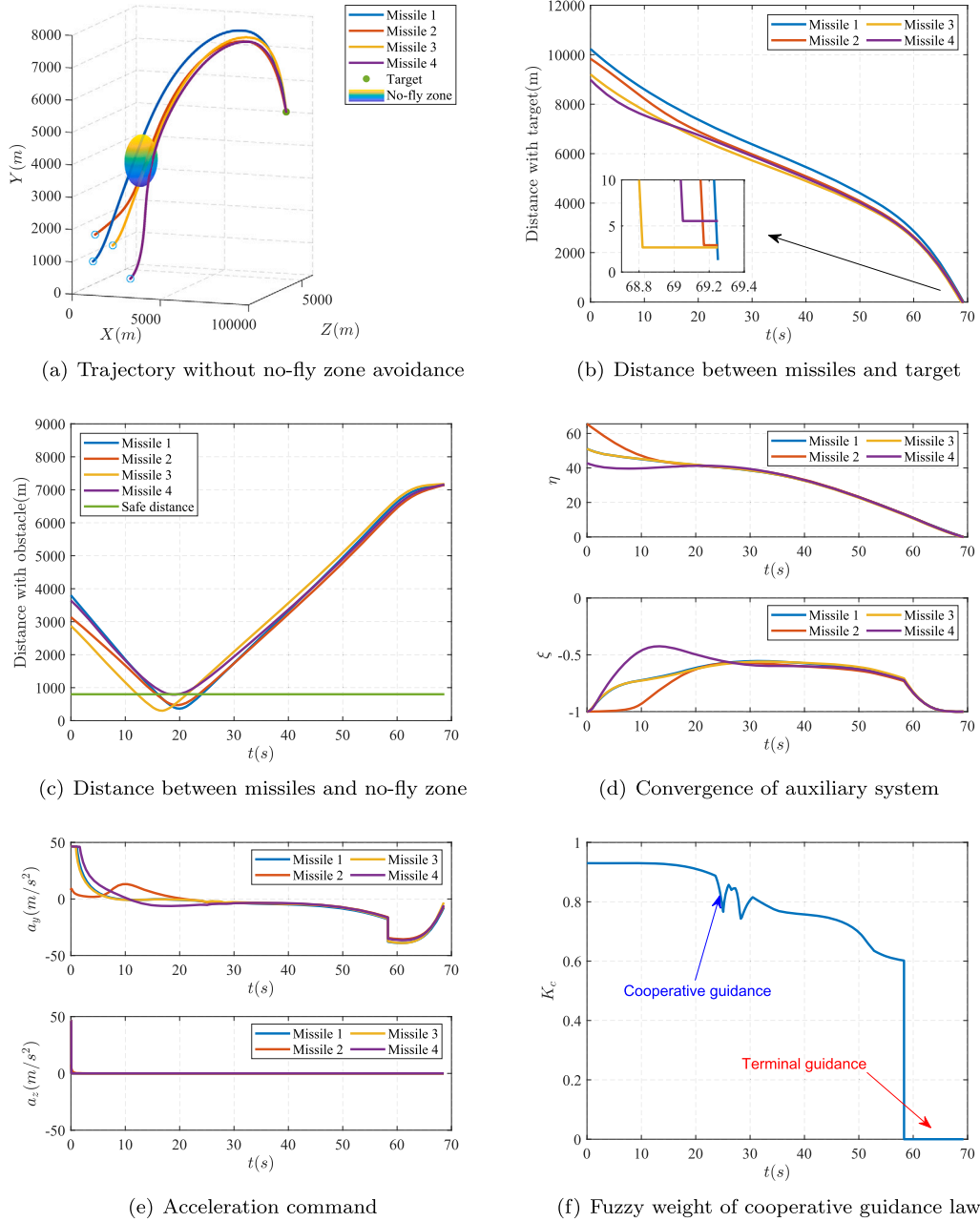


Fig. 6. Results of cooperation without no-fly zones avoidance.

the cooperative guidance law is determined by the fuzzy switching rule. The weighting coefficient for this scenario is shown in Fig. 6(f). When the missile-target distance is within 3 Km the guidance law switches to pure PNG to ensure that the leading angle converges to zero, improving terminal accuracy.

Fig. 7 shows the velocity of the aircraft. Due to the effects of gravity, thrust, and aerodynamic drag, significant variations are observed. However, during the cooperative guidance process, the acceleration in the direction of velocity is significantly smaller compared to the acceleration in the direction perpendicular to velocity, i.e., $\dot{V}_{m,i} \ll \frac{V_{m,i}^2}{r_i}$. Particularly, as the missile-target distance r_i decreases, the term $\frac{r_i \dot{V}_{m,i}}{V_{m,i}^2}$ diminishes further, allowing the velocity variation to be treated as a disturbance caused by unmodeled dynamics. Consequently, even in the presence of significant velocity changes, the existence of the closed-loop system ensures the effectiveness of the proposed guidance law.

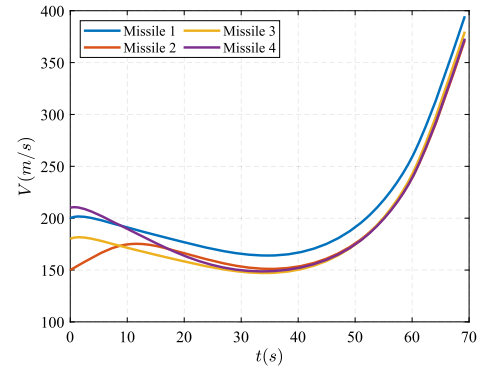


Fig. 7. Velocity of aircraft.

Table 2
Simulation initial parameters.

Item	Value
Target position (x_T, y_T, z_T)	[8500, 5000, 6000] m
Aircraft initial positions (x_0, y_0, z_0)	[1000, 1000, 300] m [1000, 1800, 500] m [2000, 1500, 500] m [3000, 500, 500] m
Aircraft initial speed v_0	[200, 150, 180, 210] m/s
No-fly zone position (x_o, y_o, z_o)	[3000, 4000, 1500] m
No-fly zone parameters p, q, r, a, b, c	1, 1, 1, 800, 800, 800
Decision distance D_{dec}	2 * 800 m
Rate coefficient k_v	0.01
Cooperative guidance law parameter k_1	0.2
PNG parameter N	3
Obstacle avoidance parameter ρ	1
Prediction step size ΔT	0.5 s
Simulation step size T_{step}	10 ms
Thrust coefficient T/m	7
Terminal distance R_a	3000 m
Convergence indicator parameters β_1, β_2	1, 50
Maximum acceleration a_y^{max}, a_z^{max}	50 m/s ² , 50 m/s ²
Boundary of leading angle $\delta_{min}, \delta_{max}$	-70°, 70°

Case 2. Cooperation with no-fly zones avoidance.

The obstacle avoidance requirement is considered by introducing an behavior based obstacle avoidance algorithm. When aircraft enters the obstacle avoidance decision area, the guidance law switches to the IFDS based obstacle avoidance algorithm. Once the aircraft exits the decision area, initial alignment is first executed, then the system transitions back to the cooperative guidance law.

Fig. 8(a) presents the 3D flight trajectories. Upon entering the obstacle avoidance decision region, the obstacles influence the initial flow field, causing the aircraft to deviate in directions away from the obstacle while maintaining orientation toward the target. After exiting the influence range of the no-fly zone and completing the initial alignment phase, the aircraft resume their trajectory toward the target under the combined effects of the cooperative guidance law and PNG.

Fig. 8(b) illustrates the missile-target distances, showing a maximum miss distance of 3.44 m. The maximum coordination error is 0.61 s, which is slightly larger than the guidance law without obstacles. This increase is attributed to the lack of cooperative behavior during the obstacle avoidance phase, which reduces the time available for synchronization among aircraft and thereby affects the cooperative outcome. Nevertheless, the final switch to PNG effectively ensures the guidance accuracy.

Fig. 8(c) illustrates the distances between the aircraft and the no-fly zone, demonstrating that the aircraft maintain a minimum safe distance of 212.03 m from the no-fly zone, effectively ensuring their safety.

Table 3
Terminal indicators in different cases.

Miss distance	Without obstacle avoidance	5.54 m
	With obstacle avoidance	3.44 m
	Monte Carlo	4.46 m
Coordination error	Without obstacle avoidance	0.43 s
	With obstacle avoidance	0.61 s
	Monte Carlo	0.75 s
Convergence time	Without obstacle avoidance	15.30 s
	With obstacle avoidance	23.46 s
	Monte Carlo	16.42 s
Safe distance	Without obstacle avoidance	- 498.39 m
	With obstacle avoidance	212.03 m
	Monte Carlo	175.14 m

Fig. 8(d) depicts the convergence behavior of the auxiliary system when the no-fly zone is considered. When switching to the behavior-based obstacle avoidance algorithm (the Flag in Fig. 8(d) is not equal to 0), the disturbed flow field ultimately converges toward the target point. As a result, the total leading angle gradually approaches zero during the obstacle avoidance phase, as shown in Fig. 8(d). Furthermore, compared to the case without obstacles, the inclusion of the no-fly zone delays the convergence time of the auxiliary variables, with the system achieving consistent convergence at $t = 23.46$ s.

Fig. 8(e) illustrates the commanded acceleration profiles and obstacle avoidance states of the missiles under no-fly zone constraints. During the initial phase of guidance, due to the presence of significant initial cooperative errors, the guidance commands experience brief saturation. Subsequently, as the cooperative errors diminish, the commands gradually decrease. When the missiles enter the no-fly zone avoidance state, the lateral acceleration a_y exhibits negative saturation, causing the missiles to avoid the no-fly zone from below, as shown in Fig. 8(a). Following this, a_y transitions to positive values, guiding the missiles toward the target. As the missiles exit the decision region, both the cooperative error and the missile-target distance become small, leading to a gradual reduction in the weighting of the cooperative guidance law, as depicted in Fig. 8(f). Fig. 9 presents the velocity profiles of the missiles. Affected by the no-fly zone, the speed of the aircraft fluctuates more violently, but good coordination results can still be obtained.

Case 3. Monte Carlo simulation.

To better evaluate the performance of the proposed algorithm, 1000 Monte Carlo simulations are conducted. The initial position and velocity are randomized within certain ranges based on the nominal trajectory presented in Case 1. Specifically, the position deviations in the X-Y axes are within [0, 500] m, the deviation in the Z direction is within [0, 100] m, and the velocity deviation is within [0, 50] m/s.

Fig. 10(a) presents the 3D trajectories in the Monte Carlo simulations. Fig. 10(b) shows the distances between the aircraft and the no-fly zone, while Fig. 10(c) illustrates the distances between the aircraft and the target. Fig. 10(d) provides the statistical results for the maximum miss distance, minimum safe distance, convergence time, and coordination error.

The median level of maximum miss distance is 4.46 m, with a maximum value of 4.99 m. The median level of minimum safety distance from the no-fly zone is 175.14 m, with a minimum value of 90.44 m. The median level of convergence time is 16.42 s, with a maximum value of 24.54 s. The median level of coordination error is 0.75 s, with a maximum value of 0.96 s. The simulation results in different cases are summarized in Table 3.

Remark 5. Considering the target is stationary, the effective operational radius of the aircraft, and the impact of obstacle avoidance on their trajectories, the maximum miss distance of 4.99 m and the maximum

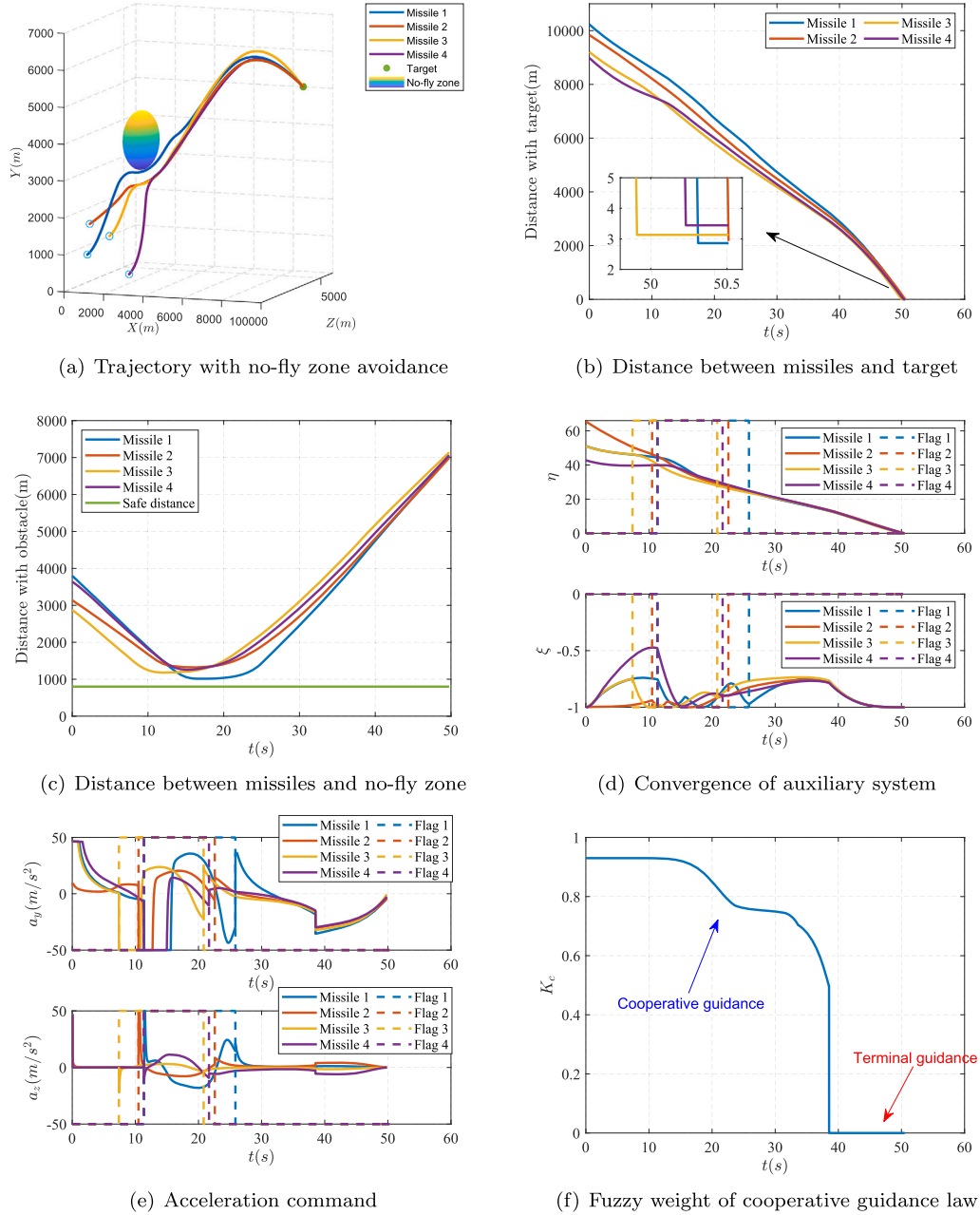


Fig. 8. Results of cooperation with no-fly zones avoidance.

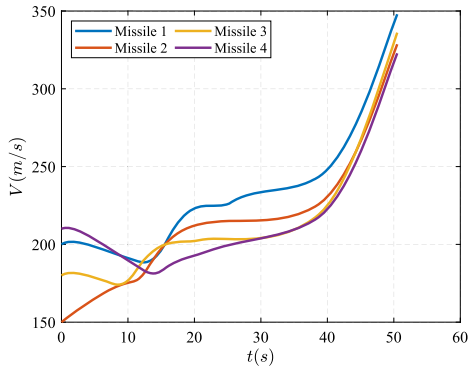


Fig. 9. Velocity of aircraft with no-fly zones avoidance.

coordination error of 0.96 s are still acceptable for actual coordinated strikes against stationary target.

The Monte Carlo simulation results demonstrate that the proposed guidance law enables effective cooperative guidance and strike, even in the presence of large initial condition deviations.

5. Conclusion

A phased cooperative guidance law is designed for aircraft without active speed control, considering FOV and input constraints, without relying on the estimated time-to-go. In the first stage, the cooperative guidance law and PNG are linearly combined based on fuzzy rules. In the second stage, the law switches entirely to PNG to ensure strike accuracy. To further enhance aircraft survivability, a behavior-based no-fly zone avoidance method is proposed. Additionally, initial alignment is introduced to ensure that the aircraft maintain FOV constraints after exiting the decision region. Extensive numerical simulations validate that, even with large initial deviations, the proposed cooperative guidance law maintains high levels of cooperative performance and strike accu-

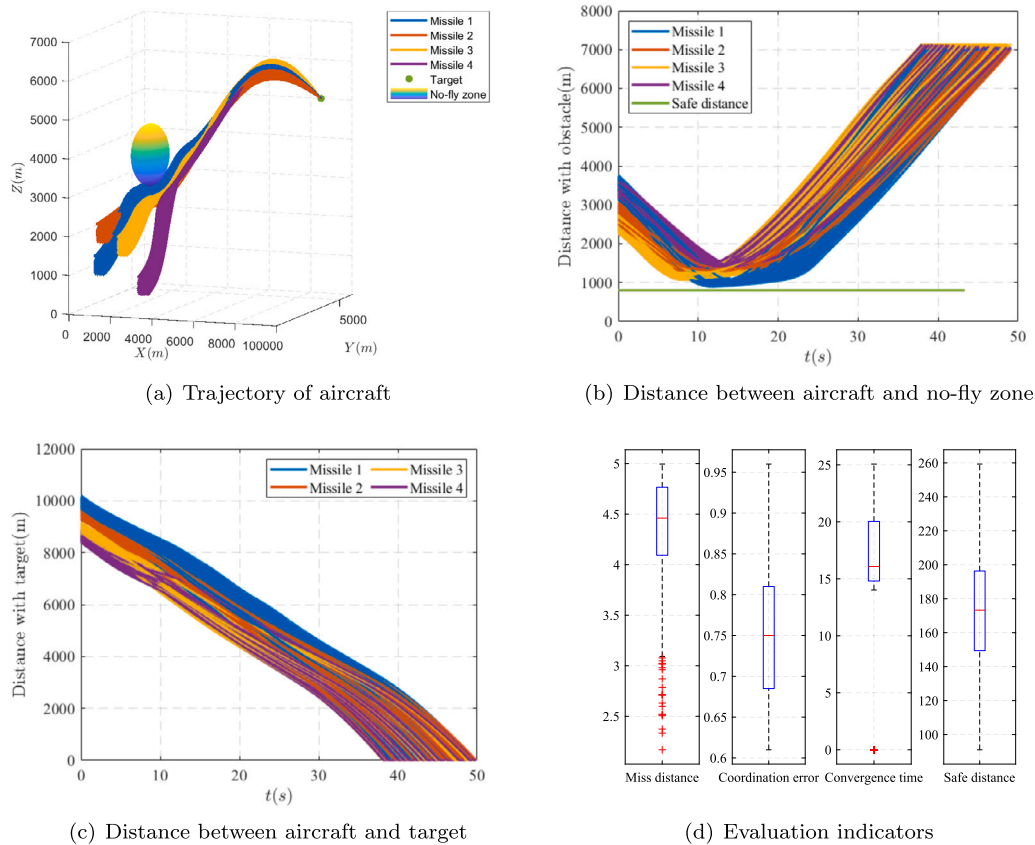


Fig. 10. Results of Monte Carlo simulation.

racy. However, the issue of internal collision avoidance among multiple aircraft is not considered during the cooperative guidance and obstacle avoidance process. As the number of vehicles increases or when cooperative guidance is implemented in small-scale scenarios, collision avoidance between vehicles becomes critically important. This area remains an open challenge and requires further comprehensive exploration.

CRediT authorship contribution statement

Kai Zhao: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology. **Jia Song:** Writing – review & editing, Writing – original draft, Funding acquisition, Formal analysis, Conceptualization. **Jingwei Yu:** Validation, Software, Methodology. **Yang Liu:** Writing – review & editing, Validation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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